

Technology Stack

Date	27 June 2026
Team ID	SWTID-2026-9346
Project Name	Voice-Based-Concept-Understanding-Analyzer
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S. No	Architecture Component/ Layer	Technology Chosen	Justification/Purpose
1	Frontend/ Client-Side (React, Angular, HTML/CSS)	Streamlit	Interactive web interface, audio upload, displaying results and dashboard
2	Backend/ Server-Side (Node.js, Python/ Django, Java)	Python, FastAPI	Python supports AI and machine learning libraries such as Whisper, Sentence-BERT, Librosa, and ReportLab. FastAPI provides high-performance REST APIs and easy integration with AI models.
3	Database / Data Storage (MySQL, MongoDB, PostgreSQL)	SQLite (Development), PostgreSQL (Production) (optional)	SQLite is lightweight and ideal for development and testing. PostgreSQL provides secure, scalable, and reliable relational data storage for user information, transcripts, and evaluation reports.
4	Cloud / Hosting / Deployment (AWS, Azure, Heroku, Docker)	Docker, Render (Backend), Vercel (Frontend)	Docker ensures consistent deployment across environments. Render provides simple backend hosting, while Vercel offers fast and reliable hosting for the React frontend.
5	Version Control & CI/CD (GitHub, GitLab, Jenkins)	Git, GitHub, GitHub Actions	Git enables version control and team collaboration. GitHub stores source code, while GitHub Actions automates testing and deployment through CI/CD pipelines.
6	Third-Party APIs/ Other Tools (Stripe, Twilio, Google Maps)	OpenAI Whisper, Google Gemini API, Sentence-Transformers, Librosa, ReportLab, Matplotlib	Whisper converts speech to text, Gemini generates AI-based feedback, Sentence-Transformers evaluates semantic similarity, Librosa extracts speech features, Matplotlib visualizes audio waveforms, and ReportLab generates downloadable PDF reports.